LEARNING NOTES

GIT & GITHUB

Exploring the tools behind version control, project history, and developer collaboration.

WHY IT MATTERS

From Java and DSA practice to future projects, Git helps organize code changes and GitHub makes collaboration possible.

A learning summary inspired by

Kunal Kushwaha’s Complete Git & GitHub Tutorial

LEARN • TRACK • COLLABORATE

01 / THE BIG PICTURE

Git and GitHub

Related tools, different roles.

GIT

A distributed version control system. It tracks changes in files and helps you manage project history on your computer.

GITHUB

A cloud-based platform for hosting Git repositories, sharing code, and collaborating with other developers.

A simple way to remember

Git manages versions. GitHub helps you host and collaborate on those versions.

Both are useful in modern software development.

02 / CORE WORKFLOW

How Git tracks your work

A common flow for recording and sharing code changes.

01 **Working Directory**

Edit or create files in your project.

02 **Staging Area**

Choose the changes you want included in the next commit.

03 **Local Repository**

Commit changes to save a snapshot in project history.

04 **Remote Repository**

Push commits to a remote repository, such as GitHub.

WORK → STAGE → COMMIT → PUSH

Commands to know

A quick reference for common Git actions.

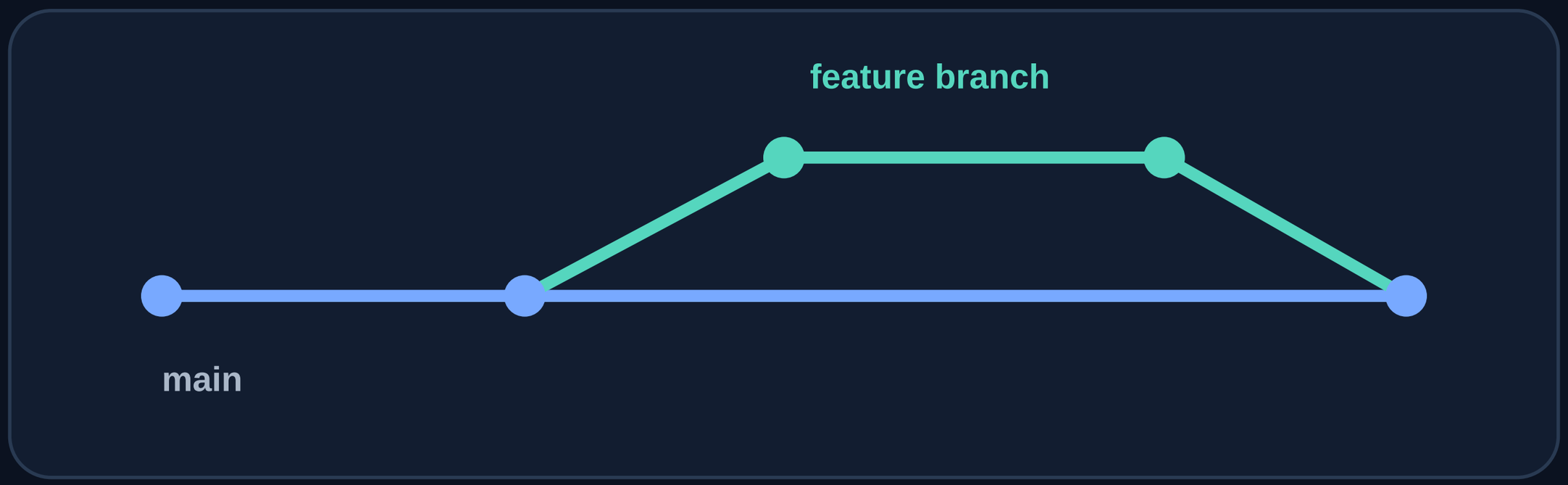
git init	Initialize a new local repository.
git status	Check the current state of your files.
git add	Stage selected changes.
git commit	Save staged changes in the local history.
git log	View commit history.
git diff	Inspect differences between versions or states.

Tip: run git status often to understand what Git sees.

04 / WORKING IN PARALLEL

Branches & merging

Branches let you work on changes without disturbing the main line of development.



- Create a branch to develop a feature or fix separately.
- Switch between branches as needed.
- Merge changes when the work is ready to integrate.

Working with others

GitHub adds collaboration workflows around Git repositories.

Fork

Create your own GitHub copy of another repository.

Clone

Download a repository to your local machine.

Remote & upstream

Connect local work to remote repositories, including the original project.

Pull Request

Propose changes and invite review before merging.

06 / BEYOND THE BASICS

More Git concepts

Useful tools for managing history and resolving changes.

Stash	Temporarily set aside uncommitted changes.
Squash	Combine multiple commits into a more compact history.
Rebase	Reapply commits on top of another base commit.
Reset	Move the current branch reference and, depending on the option, adjust the staging area or working files.
Merge conflicts	Resolve overlapping changes that Git cannot combine automatically.

TAKEAWAY

Build the habit, not just the knowledge

Version control becomes valuable when it is part of your everyday workflow.

GIT Track changes and maintain project history.

GITHUB Host repositories and collaborate with others.

NEXT Apply these concepts by managing Java and DSA practice projects with Git and GitHub.

LEARNING RESOURCE

Kunal Kushwaha — Complete Git & GitHub Tutorial (YouTube)

Learning consistently. Building practically. Growing as a developer.

#Git #GitHub #Java #DSA #SoftwareDevelopment